



UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE : WEB TECHNOLOGY AND APPLICATION
COURSE CODE : ICT600
EXAMINATION : FEBRUARY 2026
TIME : 3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts:
PART A (5 Questions)
PART B (2 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

PART A

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QUESTION 1

- a) Identify **FOUR (4)** functions that an author can do with HTML. (8 marks)
- b) List **FIVE (5)** popular web-based Content Management System (CMS) tools commonly used for creating and managing websites or blogs. (10 marks)

QUESTION 2

- a) Describe briefly **SIX (6)** key components of website anatomy. (6 marks)
- b) Name **SIX (6)** development tools that can be used in client-side web development. (6 marks)

QUESTION 3

- a) Prepare a PHP script to create a database in MySQL including the SQL statement, execution, error handling, and closing of the connection. (8 marks)
- b) From the web page example as shown in Figure 1, produce a PHP file (e.g. add.php) to add a new user to the database.

The figure shows two side-by-side screenshots of a web form titled 'User Entry'.
The left screenshot displays the form with four input fields: 'Username', 'Password', 'Role', and 'Status'. Below these fields are two buttons: 'OK' and 'Back'.
The right screenshot shows the same form after a successful submission. A message 'New record created successfully' is displayed below the input fields. The 'OK' and 'Back' buttons are still present at the bottom.

Figure 1 : User Entry

(10 marks)

QUESTION 4

The Model-View-Controller (MVC) architecture is a widely adopted design pattern in web development, as it facilitates the separation of application components, thereby enhancing code clarity, maintainability, and accessibility, even for novice developers.

- a) Based on the following partial CodeIgniter controller class, construct the index method for the *Employee* controller. The index method will load and display the *employee_profile.php* view once the controller is executed.

```
?>class Employee extends CI_Controller {  
    // write the index method after this line.  
}  
?>
```

(4 marks)

- b) Illustrate an MVC architecture diagram that demonstrates the process in which the web browser initiates a request to the *Employee* controller, and the controller subsequently loads and displays the *Employee Profile* view.

(6 marks)

QUESTION 5

The OWASP Top 10 is a widely recognized awareness document that highlights the most critical security risks to web applications, providing developers with a consensus-based guideline for improving application security.

- a) Assess the risk and **TWO (2)** countermeasures when data is transmitted in clear text between a web browser and a web server.

(6 marks)

- b) Assess the risk and **TWO (2)** countermeasures when a web page contains a form requesting the user to upload a file to the web application.

(6 marks)

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PART B

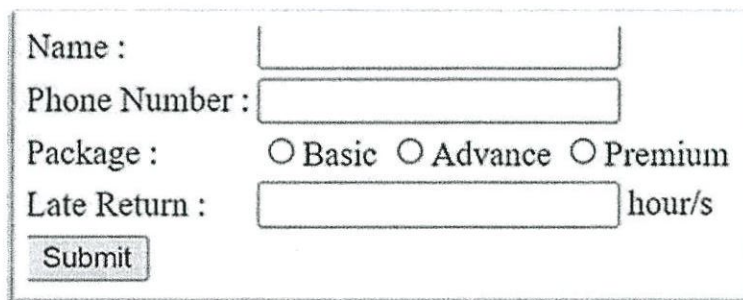
QUESTION 1

You are assigned to develop a PHP-based web page for *sewaCompany*, a business that provides bicycle rentals for visitors engaging in recreational and leisure activities. The charges for renting bicycle are shown in the Table 1.

Table 1: Bicycle Renting Charges

Package type	Duration	Charge	Late Charge
Basic(B)	1 Hour	RM15	Additional RM10 for every hour
Advanced(A)	3 Hours	RM23	
Premium(P)	Unlimited	RM100	-

You are required to develop **TWO (2)** pages, *inputData.html* and *calculate.php*. The *inputData.html* page should contain the input fields as illustrated in Figure 2. Once user clicks submit button, the data must be sent to the *calculate.php* page. Write a PHP program to calculate the total rental charge the customer must pay and display the output in the format as shown in Figure 3.



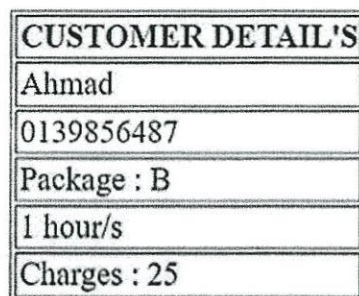
Name :

Phone Number :

Package : ☐ Basic ☐ Advance ☐ Premium

Late Return : hour/s

Figure 2: inputData.html



CUSTOMER DETAIL'S
Ahmad
0139856487
Package : B
1 hour/s
Charges : 25

Figure 3: calculate.php

(15 marks)

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QUESTION 2

You are required to interpret a page into JavaScript code that contains a form for user to input a number as illustrated in the Figure 4. For the first page user enter a number and click button "Send", the page-should display the number entered, the maximum number (the number user entered) and the total number (also the number user entered). Subsequently, as the user continues to input numbers, the page (as shown in Figure 5) should dynamically update to display:

- all numbers user had entered,
- the current maximum value among the numbers entered
- the cumulative total for all numbers entered.



Number:

The value you entered:

The maximum number you entered:

The total number you entered:

Figure 4: First Page

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Number: 2 Send The value you entered: 2 The maximum number you entered: 2 The total number you entered: 2	Number: 5 Send The value you entered: 2 The maximum number you entered: 2 The total number you entered: 5	Number: 4 Send The value you entered: 2,5 The maximum number you entered: 5 The total number you entered: 7
Number: 15 Send The value you entered: 2,5,4 The maximum number you entered: 5 The total number you entered: 11	Number: Send The value you entered: 2,5,4,15 The maximum number you entered: 15 The total number you entered: 26	

Figure 5: Series of Input

(15 marks)

END OF QUESTION PAPER

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